

Smallest enclosing balls by an active-set method on the dual

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Abstract

We record the dual quadratic program of the smallest enclosing ball problem, the active-set method that solves it, and the invariance properties an implementation must preserve. The method terminates at an exact vertex of the dual feasible set rather than at an interior-point tolerance, costs one $k \times k$ dense solve per iteration with $k \leq d$, and returns the dual weights, which are the conic dual of the second-order cone formulation after one rescaling and hence a certificate the caller can check independently.

1 The problem and its dual

For $S = \{p_1, \dots, p_n\} \subset \mathbb{R}^d$ the smallest enclosing ball is the solution of

$$\min_{x, r} r \quad \text{s.t.} \quad \|p_i - x\| \leq r, \quad i = 1, \dots, n, \quad (1)$$

a second-order cone program in $(r, x) \in \mathbb{R}^{1+d}$ with n blocks $(r, p_i - x) \in \mathcal{Q}^{d+1}$. Existence follows from coercivity of $f(x) = \max_i \|p_i - x\|$ and uniqueness from strict convexity of f^2 along any segment; we write (R, x^*) for the optimum.

Let $\Delta_n = \{u \in \mathbb{R}^n : u \geq 0, \mathbf{1}^\top u = 1\}$, and for $u \in \Delta_n$ put $x(u) = \sum_i u_i p_i$ and

$$\varphi(u) = \sum_i u_i \|p_i\|^2 - \|x(u)\|^2, \quad (2)$$

a concave quadratic. Everything below rests on one identity, valid for all $y \in \mathbb{R}^d$ and $u \in \Delta_n$:

$$\sum_i u_i \|p_i - y\|^2 = \varphi(u) + \|x(u) - y\|^2. \quad (3)$$

Proposition 1. $R^2 = \max_{u \in \Delta_n} \varphi(u)$, and $x^* = x(w)$ for any maximiser w .

Proof. If $B(y, r) \supseteq S$ then the left side of (3) is at most r^2 , so $\varphi(u) \leq r^2$ for every u ; taking the optimum gives $\max \varphi \leq R^2$. Conversely let $T = \{p_i : \|p_i - x^*\| = R\}$. If $x^* \notin \text{conv } T$, separate: there is v with $v^\top(p - x^*) > 0$ on T , and $\|p - (x^* + tv)\|^2 = R^2 - 2tv^\top(p - x^*) + t^2\|v\|^2 < R^2$ for small $t > 0$, while the finitely many points off T stay strictly inside — so the radius decreases, a contradiction. Hence $x^* = \sum_i w_i p_i$ for some $w \in \Delta_n$ supported on T , and $y = x^*$ in (3) gives $\varphi(w) = \sum_i w_i \|p_i - x^*\|^2 = R^2$. The same substitution forces $\|x(w) - x^*\|^2 \leq 0$ for any maximiser. $\square$

Since $\partial\varphi/\partial u_i = \|p_i - x(u)\|^2 - \|x(u)\|^2$, the first-order conditions for $\max_{\Delta_n} \varphi$ read, after cancelling the common term,

$$\|p_i - x\| = R \quad (w_i > 0), \quad \|p_i - x\| \leq R \quad (w_i = 0), \quad (4)$$

the geometric certificate: the points carrying weight lie on the boundary sphere and the rest are enclosed. By Carathéodory the support may be taken of size at most $d + 1$, so (1) is determined by at most $d + 1$ of the n points and the problem is combinatorial in the choice of that subset.

2 The method

Maintain a working set F and weights $u > 0$ on it summing to one.

Subproblem. Maximising φ over weights supported on F with no sign restriction is, by (4) without the inequalities, the requirement that x be equidistant from $\{q_j\}_{j \in F}$ and lie in their affine hull. With D the $(m - 1) \times d$ matrix of edges $e_j = q_j - q_0$ and $x = q_0 + D^\top y$,

$$(DD^\top)y = \frac{1}{2}(\|e_1\|^2, \dots, \|e_{m-1}\|^2)^\top, \quad w = \left(1 - \sum_j y_j, y\right), \quad (5)$$

a dense system of order $m - 1 \leq d$, non-singular exactly when F is affinely independent. This is the circumcentre of the face, expressed in barycentric coordinates.

Dropping. If $\min_j w_j < 0$ the circumcentre lies outside $\text{conv}\{q_j\}$ and the subproblem solution is dual-infeasible. Move along $s = w - u$ by the ratio test

$$\alpha = \min\{-u_i/s_i : s_i < 0\}, \quad (6)$$

which drives at least one weight to zero; remove it from F .

Degeneracy. If F is affinely dependent, $DD^\top$ is singular. The directions s with $\mathbf{1}^\top s = 0$ and $\sum_i s_i q_i = 0$ — equivalently $D^\top s_{\setminus 1} = 0$ after eliminating s_0 — leave $x(u)$ fixed, so $\|x(u)\|^2$ is frozen and $\varphi(u + ts) = \varphi(u) + t \sum_i s_i \|p_i\|^2$ is linear: the subproblem is unbounded. Project the gradient onto that null space and apply (6), shrinking F by one.

Adding. Otherwise accept $u \leftarrow w$, set $x = x(u)$ and $R = \max_{i \in F} \|p_i - x\|$, and let $k = \arg \max_i \|p_i - x\|$. If $\|p_k - x\| \leq R$ then (4) holds and (R, x, u) is optimal; otherwise append k to F with weight zero. By (7) below this is also the steepest-ascent choice among the fixed variables.

Input $p_1, \dots, p_n$. Recentre on $\bar{p}$. Set $F = \{k\}$, k farthest from p_1 ; $u_k = 1$.
Repeat: if F affinely dependent, take the null-space descent step and drop; else solve (5). If $\min w < 0$, take the ratio step (6) and drop. Else set $u = w$, $x = x(u)$, $R = \max_{i \in F} \|p_i - x\|$; if no point exceeds R , return $(R, x + \bar{p}, u)$; else drop zero weights and append $\arg \max_i \|p_i - x\|$ with weight 0.

Each iteration either strictly increases φ or strictly shrinks F , and there are finitely many working sets, so the method is finite; as with any active-set method a degenerate zero-length step requires an anti-cycling rule in exact arithmetic, and in practice an iteration cap serves as a safety net. Note that

$$\partial\varphi/\partial u_k - \partial\varphi/\partial u_i = \|p_k - x\|^2 - \|p_i - x\|^2, \quad (7)$$

so the farthest violator is the fixed variable with the largest reduced gradient.

3 Invariance

Two properties are load-bearing and easily lost. Both are consequences of refusing any tolerance tied to coordinates of magnitude one.

Origin invariance. Recentre on $\bar{p}$ before iterating, and form $\|p - x\|^2$ as $\sum_k (p_k - x_k)^2$ rather than $\|p\|^2 - 2p^\top x + \|x\|^2$. The expanded form cancels quantities of size $\|x\|^2$ to produce an answer of size R^2 , losing every digit by which $\|x\|$ exceeds R ; for a cloud of unit extent at distance 10^6 the rounding floor of a squared distance is then of the order of the radius itself.

Scale invariance. Test affine dependence on the rank of D , not on $[\mathbf{1}; Q^\top]$: the row of ones has unit entries, dominates the singular values, and would declare any cloud of extent below ε degenerate. Size the feasibility slack of the stopping test as a relative tolerance plus an absolute floor proportional to $\varepsilon \max_i \|p_i\|^2$, and normalise the null-space direction before comparing anything derived from it — the gradient carries units of length², the weight tolerance does not.

4 Cost, and the dual as a certificate

Per iteration the method performs one $O(nd)$ sweep of squared distances and one dense solve of order at most d ; nothing of size n is ever factorised. This is the structural difference from an interior-point method applied to (1), which factors an $n(d+1)$ -row system at every step. The method also identifies the support combinatorially and solves that face exactly, rather than approaching it from the interior and leaving the caller to decide which slacks of size 10^{-9} were meant to be zero.

The weights are the conic dual of (1) after one rescaling: the block for point i is

$$z_i = u_i \left(1, -\frac{p_i - x}{R} \right) \in \mathcal{Q}^{d+1}, \quad (8)$$

whose tail has norm $u_i \|p_i - x\|/R$, equal to its head exactly on the support, and which pairs with the slack $s_i = (R, p_i - x)$ as $s_i^\top z_i = (u_i/R)(R^2 - \|p_i - x\|^2) = 0$ — complementarity, each factor vanishing where the other does not. So a solver built on this method returns a standard-form primal–dual pair, and a caller can verify optimality from (4) at the cost of one pass over the data, without re-solving.

Remark (Substitutability). This is what makes the method usable behind a conic solver’s interface rather than only as a special-purpose routine. The accompanying implementation dispatches on the cone list: zero and non-negative blocks go to a dense dual active-set QP method [4], and equally sized second-order blocks matching the shape of (1) go to the method above. A program carrying the right cones but a different matrix is refused rather than answered, which requires reconstructing the point cloud from (A, b) and rebuilding the program for comparison. An indefinite objective is likewise refused rather than regularised into definiteness, since $P + \epsilon I$ answers a different question.

5 A numerical comparison

Table 1 compares the method with Clarabel 0.11.1 applied to (1) on Gaussian clouds, measuring the relative enclosure error $\max_i \|p_i - x\|/R - 1$ — positive values mean the returned ball fails to contain the cloud. Both agree on R to the digits shown. The active-set figures are 0 or a single ulp, since the returned R is by construction the exact maximum distance over the support of a centre computed from (5); Clarabel’s are its interior-point residuals, four to five orders larger. The timing gap grows with n because only the $O(nd)$ sweep depends on it.

n	d	enclosure error		time (s)	
		active set	Clarabel	active set	Clarabel
10^3	2	0	$5.3 \cdot 10^{-12}$	0.001	0.009
10^3	10	0	$9.4 \cdot 10^{-12}$	0.001	0.056
10^4	3	$2.2 \cdot 10^{-16}$	$2.0 \cdot 10^{-12}$	0.001	0.141
10^4	20	$-1.1 \cdot 10^{-16}$	$3.5 \cdot 10^{-14}$	0.006	0.948
10^5	3	$-2.2 \cdot 10^{-16}$	$1.7 \cdot 10^{-11}$	0.008	2.225
10^5	10	0	$-7.2 \cdot 10^{-12}$	0.014	6.595

Table 1: Standard normal clouds, one seed per row; Apple M4 Pro, CPython 3.12.13, NumPy 2.5.1, Clarabel 0.11.1 at its default tolerances. Times are a single run and are indicative only; the enclosure errors are not, being deterministic properties of the returned pair.

The comparison is not entirely fair to Clarabel, which is built for large sparse programs of arbitrary shape and is here being asked for a dense problem of one particular shape whose combinatorial structure the other method exploits. That is the point: the specialised method is worth keeping precisely where the structure is known.

6 Relation to the minimum-norm point problem

If all points have equal norm ρ — for instance after the standard lift of a kernel problem onto a sphere — then (2) reduces to $\varphi(u) = \rho^2 - \|x(u)\|^2$, so maximising φ over Δ_n is minimising $\|x\|$ over $\text{conv } S$: the minimum-norm point problem solved by Wolfe’s algorithm and by the Frank–Wolfe family. The working set of Section 2 is then Wolfe’s corral, the subproblem (5) is his affine minimiser, and the drop step is his major cycle. The general case is the same algorithm with the linear term $\sum_i u_i \|p_i\|^2$ restored, which is what makes the subproblem a circumcentre rather than a projection.

7 Notes

The problem is Sylvester’s [6]; Elzinga and Hearn [1] gave an early finite algorithm of this type. Welzl’s randomised incremental method [8] is expected $O(n)$ for fixed d but degrades in d ; Gärtner [3] made it robust, and Fischer, Gärtner and Kutz [2] developed the support-set framework the method above belongs to, including the balls-of-balls generalisation. The QP machinery is standard [5]. An implementation of the method, of the Goldfarb–Idnani core, and of the conic front end is available at <https://github.com/tschm/cvxball>.

References

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